# PG No. 13 $C_{4v}$ $4mm$ [ tetragonal ] (internal axial dipole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(1,1)}[g](A_2)$

** symmetry

1

** expression

$$\frac{\sqrt{3}G_x x}{3} + \frac{\sqrt{3}G_y y}{3} + \frac{\sqrt{3}G_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{Q}}_1^{(1,0)}[g](A_1)$

** symmetry

$z$

** expression

$$\frac{\sqrt{2}G_x y}{2} - \frac{\sqrt{2}G_y x}{2}$$

$\vec{\mathbb{G}}_1^{(1,-1)}[g](A_2)$

** symmetry

$z$

** expression

$G_z$

$\vec{\mathbb{G}}_1^{(1,1)}[g](A_2)$

** symmetry

$z$

** expression

$$\frac{3\sqrt{10}G_x xz}{10} + \frac{3\sqrt{10}G_y yz}{10} - \frac{\sqrt{10}G_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_{1,1}^{(1,0)}[g](E)$, $\vec{\mathbb{Q}}_{1,2}^{(1,0)}[g](E)$

** symmetry

$x$

$y$

** expression

$$\frac{\sqrt{2}G_y z}{2} - \frac{\sqrt{2}G_z y}{2}$$

$$-\frac{\sqrt{2}G_x z}{2} + \frac{\sqrt{2}G_z x}{2}$$

$\vec{\mathbb{G}}_{1,1}^{(1,-1)}[g](E)$, $\vec{\mathbb{G}}_{1,2}^{(1,-1)}[g](E)$

** symmetry

$-y$

$x$

** expression

$-G_y$

$G_x$

$\vec{\mathbb{G}}_{1,1}^{(1,1)}[g](E)$, $\vec{\mathbb{G}}_{1,2}^{(1,1)}[g](E)$

** symmetry

$-y$

$x$

** expression

$$-\frac{3\sqrt{10}G_x xy}{10} + \frac{\sqrt{10}G_y\left(x^2-2y^2+z^2\right)}{10} - \frac{3\sqrt{10}G_z yz}{10}$$

$$\frac{\sqrt{10}G_x\left(2x^2-y^2-z^2\right)}{10}+\frac{3\sqrt{10}G_yxy}{10}+\frac{3\sqrt{10}G_zxz}{10}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(1,0)}[g](A_1)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{6}G_xyz}{2}-\frac{\sqrt{6}G_yxz}{2}$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](A_2)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{6}G_xx}{6}-\frac{\sqrt{6}G_yy}{6}+\frac{\sqrt{6}G_zz}{3}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](A_2)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}G_xx\left(x^2+y^2-4z^2\right)}{14}-\frac{\sqrt{21}G_yy\left(x^2+y^2-4z^2\right)}{14}-\frac{\sqrt{21}G_zz\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(1,0)}[g](B_1)$

** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{2}G_xyz}{2}+\frac{\sqrt{2}G_yxz}{2}-\sqrt{2}G_zxy$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](B_1)$

** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}G_xy}{2}+\frac{\sqrt{2}G_yx}{2}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](B_1)$

** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{7}G_xy\left(4x^2-y^2-z^2\right)}{7}-\frac{\sqrt{7}G_yx\left(x^2-4y^2+z^2\right)}{7}+\frac{5\sqrt{7}G_zxyz}{7}$$

$\vec{\mathbb{Q}}_2^{(1,0)}[g](B_2)$

** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{2}G_xxz}{2}+\frac{\sqrt{2}G_yyz}{2}+\frac{\sqrt{2}G_z\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](B_2)$

** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{2}G_x x}{2} - \frac{\sqrt{2}G_y y}{2}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](B_2)$
** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{7}G_x x\left(3x^2 - 7y^2 - 2z^2\right)}{14} + \frac{\sqrt{7}G_y y\left(7x^2 - 3y^2 + 2z^2\right)}{14} + \frac{5\sqrt{7}G_z z\left(x - y\right)\left(x + y\right)}{14}$$

$\vec{\mathbb{Q}}_{2,1}^{(1,0)}[g](E)$, $\vec{\mathbb{Q}}_{2,2}^{(1,0)}[g](E)$
** symmetry

$$\sqrt{3}xz$$

$$\sqrt{3}yz$$

** expression

$$\frac{\sqrt{2}G_x xy}{2} - \frac{\sqrt{2}G_y\left(x - z\right)\left(x + z\right)}{2} - \frac{\sqrt{2}G_z yz}{2}$$

$$\frac{\sqrt{2}G_x\left(y - z\right)\left(y + z\right)}{2} - \frac{\sqrt{2}G_y xy}{2} + \frac{\sqrt{2}G_z xz}{2}$$

$\vec{\mathbb{G}}_{2,1}^{(1,-1)}[g](E)$, $\vec{\mathbb{G}}_{2,2}^{(1,-1)}[g](E)$
** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$\frac{\sqrt{2}G_y z}{2} + \frac{\sqrt{2}G_z y}{2}$$

$$-\frac{\sqrt{2}G_x z}{2} - \frac{\sqrt{2}G_z x}{2}$$

$\vec{\mathbb{G}}_{2,1}^{(1,1)}[g](E)$, $\vec{\mathbb{G}}_{2,2}^{(1,1)}[g](E)$
** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$\frac{5\sqrt{7}G_x xyz}{7} - \frac{\sqrt{7}G_y z\left(x^2 - 4y^2 + z^2\right)}{7} - \frac{\sqrt{7}G_z y\left(x^2 + y^2 - 4z^2\right)}{7}$$

$$-\frac{\sqrt{7}G_x z\left(4x^2 - y^2 - z^2\right)}{7} - \frac{5\sqrt{7}G_y xyz}{7} + \frac{\sqrt{7}G_z x\left(x^2 + y^2 - 4z^2\right)}{7}$$

* Harmonics for rank 3
$\vec{\mathbb{Q}}_3^{(1,0)}[g](A_1)$
** symmetry

$$-\frac{3x^2 z}{2} - \frac{3y^2 z}{2} + z^3$$

** expression

$$-\frac{\sqrt{3}G_x y\left(x^2 + y^2 - 4z^2\right)}{4} + \frac{\sqrt{3}G_y x\left(x^2 + y^2 - 4z^2\right)}{4}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](A_2)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}G_xxz}{5}-\frac{\sqrt{15}G_yyz}{5}-\frac{\sqrt{15}G_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](A_2)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{5G_xxz\left(3x^2+3y^2-4z^2\right)}{12}-\frac{5G_yyz\left(3x^2+3y^2-4z^2\right)}{12}+\frac{G_z\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](B_1)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{5}G_xy\left(x^2-y^2+2z^2\right)}{4}-\frac{\sqrt{5}G_yx\left(x^2-y^2-2z^2\right)}{4}-\sqrt{5}G_zxyz$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](B_1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$G_xyz+G_yxz+G_zxy$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](B_1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{15}G_xyz\left(6x^2-y^2-z^2\right)}{6}-\frac{\sqrt{15}G_yxz\left(x^2-6y^2+z^2\right)}{6}-\frac{\sqrt{15}G_zxy\left(x^2+y^2-6z^2\right)}{6}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](B_2)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{5}G_xx\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{5}G_yy\left(x-z\right)\left(x+z\right)}{2}+\frac{\sqrt{5}G_zz\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](B_2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$G_xxz-G_yyz+\frac{G_z\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](B_2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{15}G_xxz\left(5x^2-9y^2-2z^2\right)}{12}+\frac{\sqrt{15}G_yyz\left(9x^2-5y^2+2z^2\right)}{12}-\frac{\sqrt{15}G_z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12}$$

$\vec{\mathbb{Q}}^{(1,0)}_{3,1}[g](E,1)$, $\vec{\mathbb{Q}}^{(1,0)}_{3,2}[g](E,1)$
** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

** expression

$$\frac{\sqrt{3}G_yz\left(4x^2 - y^2 - z^2\right)}{4} - \frac{\sqrt{3}G_zy\left(4x^2 - y^2 - z^2\right)}{4}$$

$$\frac{\sqrt{3}G_xz\left(x^2 - 4y^2 + z^2\right)}{4} - \frac{\sqrt{3}G_zx\left(x^2 - 4y^2 + z^2\right)}{4}$$

$\vec{\mathbb{Q}}^{(1,0)}_{3,1}[g](E,2)$, $\vec{\mathbb{Q}}^{(1,0)}_{3,2}[g](E,2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$\frac{\sqrt{15}x^2y}{2} - \frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\sqrt{5}G_xxyz + \frac{\sqrt{5}G_yz\left(2x^2 + y^2 - z^2\right)}{4} + \frac{\sqrt{5}G_zy\left(2x^2 - y^2 + z^2\right)}{4}$$

$$-\frac{\sqrt{5}G_xz\left(x^2 + 2y^2 - z^2\right)}{4} + \sqrt{5}G_yxyz + \frac{\sqrt{5}G_zx\left(x^2 - 2y^2 - z^2\right)}{4}$$

$\vec{\mathbb{G}}^{(1,-1)}_{3,1}[g](E,1)$, $\vec{\mathbb{G}}^{(1,-1)}_{3,2}[g](E,1)$
** symmetry

$$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$$

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

** expression

$$\frac{\sqrt{15}G_xxy}{5} + \frac{\sqrt{15}G_y\left(x^2 - 2y^2 + z^2\right)}{10} + \frac{\sqrt{15}G_zyz}{5}$$

$$\frac{\sqrt{15}G_x\left(2x^2 - y^2 - z^2\right)}{10} - \frac{\sqrt{15}G_yxy}{5} - \frac{\sqrt{15}G_zxz}{5}$$

$\vec{\mathbb{G}}^{(1,-1)}_{3,1}[g](E,2)$, $\vec{\mathbb{G}}^{(1,-1)}_{3,2}[g](E,2)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

** expression

$$-G_xxy - \frac{G_y\left(x - z\right)\left(x + z\right)}{2} + G_zyz$$

$$\frac{G_x\left(y - z\right)\left(y + z\right)}{2} + G_yxy - G_zxz$$

$\vec{\mathbb{G}}^{(1,1)}_{3,1}[g](E,1)$, $\vec{\mathbb{G}}^{(1,1)}_{3,2}[g](E,1)$
** symmetry

$$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$$

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

** expression

$$\frac{5G_x xy\left(3x^2-4y^2+3z^2\right)}{12}-\frac{G_y\left(3x^4-24x^2y^2+6x^2z^2+8y^4-24y^2z^2+3z^4\right)}{12}+\frac{5G_z yz\left(3x^2-4y^2+3z^2\right)}{12}$$

$$\frac{G_x\left(8x^4-24x^2y^2-24x^2z^2+3y^4+6y^2z^2+3z^4\right)}{12}+\frac{5G_y xy\left(4x^2-3y^2-3z^2\right)}{12}+\frac{5G_z xz\left(4x^2-3y^2-3z^2\right)}{12}$$

$\vec{\mathbb{G}}_{3,1}^{(1,1)}[g](E,2)$, $\vec{\mathbb{G}}_{3,2}^{(1,1)}[g](E,2)$

** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

** expression

$$-\frac{\sqrt{15}G_x xy\left(5x^2-2y^2-9z^2\right)}{12}+\frac{\sqrt{15}G_y\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{12}-\frac{\sqrt{15}G_z yz\left(9x^2+2y^2-5z^2\right)}{12}$$

$$\frac{\sqrt{15}G_x\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{12}-\frac{\sqrt{15}G_y xy\left(2x^2-5y^2+9z^2\right)}{12}+\frac{\sqrt{15}G_z xz\left(2x^2+9y^2-5z^2\right)}{12}$$